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Preliminary palaeomagnetic results from the Subbetic Zone (Betic Cordillera, southern Spain): kinematic and structural implications

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We present palaeomagnetic evidence from Jurassic limestones and basic submarine volcanic rocks (sills, pillow lavas, and intrusive material) from the central part of the Subbetic Zone of the Betic Cordillera. Isothermal remanent magnetization acquisition experiments performed on limestones and thermomagnetic curves and measurements of the magnetic susceptibility determined for the volcanic rocks demonstrate strong dependence of rock magnetic properties on lithology. In most cases a stable magnetization could be isolated after thermal and/or alternating field demagnetization. Locally, declinations of the characteristic remanent magnetization from interbedded limestones and volcanics are similar, but inclinations tend to vary. Regionally, declinations for both limestones and volcanics show significant clockwise rotation. These results cannot be explained by the anticlockwise rotation of Iberia during the Cretaceous opening of the Bay of Biscay. Considering the long and complex history of large-scale shear between Africa and Europe accommodated in southern Spain, the anomalous declinations can be explained by a model with rotation of crustal blocks within a zone of wrench tectonics.

1. Introduction

The Betic Cordillera can be divided into three zones (Fig. 1), the most external, or Prebetic Zone mostly northeast of Granada, the Subbetic Zone in the middle (itself differentiated into external, middle, and internal zones), and the internal, partly metamorphic Betic Zone to the south. The Subbetic Zone comprises platform, hemipelagic, and pelagic sediments and submarine volcanics deposited or extruded on the southern margin of the Iberian Plate during the Mesozoic break-up of Pangea. Vegas and Muñoz (1984) have ascribed the Jurassic sedimentation of the Subbetic Zone to horst–graben tectonics related to the oblique extension between Africa and Iberia. According to their model, the widespread Jurassic submarine volcanic activity of the Subbetic Zone can be related to a broad system of volcanic outpourings that occurred during the peak of the Jurassic transtensional event along the entire transform zone between the Iberian Chain of eastern Spain (Vegas, 1985) and the Atlas System in southern Morocco (Jenny, 1984).

The southern border of the Iberian Plate, the site of the Betic orogeny, has been affected by transcurrent movements since the differential separation of the North and South Atlantic beginning in the Mesozoic and continuing to the present (Paquet, 1972; Araña and Vegas, 1974; Hermes, 1978; Bousquet, 1979; Van de Fliert et al., 1980; Vegas et al., 1980; De Smet, 1984; Leblanc and Olivier, 1984; Dercourt et al., 1986). However, the role, magnitude, and sense of movement of blocks in this region are poorly defined. In the past, geologists have tended to place greater emphasis

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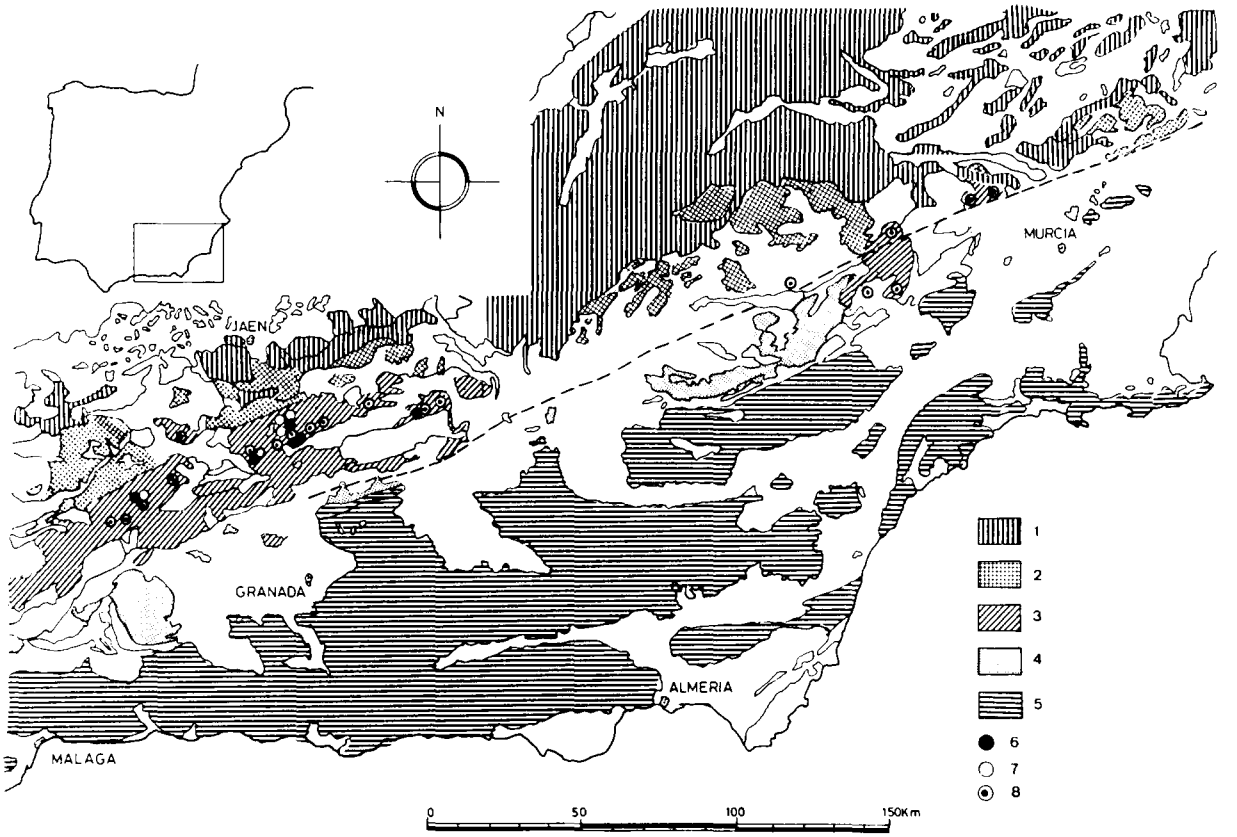


Fig. 1. Geological map of the distribution at the External and International Zones of the Betic Cordilleras (simplified from Azema et al. (1979)) and location of the sampled sites. External Zones (1–4): (1) Prebetic, (2) External Subbetic, (3) Middle Subbetic, (4) Internal Subbetic, (5) Internal Zones including the 'Dorsale Calcaire', (6) palaeomagnetic sites in volcanic rocks, (7) palaeomagnetic sites in limestones, (8) Jurassic volcanism. The broken line is the main trace of the Crevillente Fault; we have not extended this line west of Granada since the exact path is uncertain, although the fault itself certainly continues westwards.

on N–S compression rather than on E–W translation. Thus, until the 1980s, most geologic structures were interpreted within the context of thrust tectonics. Very probably, this zone has been the scene of complex transpression involving both sinistral and dextral wrenching, N–S compression, and even Neogene extension (Vegas et al., 1980).

Palaeomagnetic declinations are very sensitive indicators of rotations about vertical axes, movements which could be expected of blocks within a zone which has undergone transcurrent faulting. However, published palaeomagnetic data from the Subbetic Zone (Vandenberg, 1980; Mäkel et al., 1984; Ogg et al., 1984; Steiner et al., 1987) are too sparse to use as tectonic indicators. Therefore we initiated a regional palaeomagnetic study in both limestones and volcanic rocks of the Subbetic Zone to supply additional data to help define and quantify the complex tectonics of this region.

2. Geological setting and description of the sampling localities

We investigated 11 sites from five localities in the central part of the Subbetic Zone in both limestones and submarine volcanic rocks of Jurassic age. At first sight, the tectonic situation looks deceptively simple (from a palaeomagnetic point of view) with very little penetrative deforma-

tion or stylolite formation. This is probably due to the unique circumstances of the Subbetic Zone where most of the deformation is taken up by Triassic marls and evaporites which provide an ideal lubricating medium (De Smet, 1984). On more careful inspection, however, local faulting may have been quite complex, involving reactivation of Palaeozoic tectonic features during Mesozoic transtension and Cenozoic transpression, and initiation of new tectonic features in the Neogene and Quaternary. Perhaps the most significant development is the large amount of strike-slip faulting that has occurred during and since the Oligocene–Miocene orogeny. This re-evaluation of local tectonics has led us to withhold the interpretation of some of the sites, even though the palaeomagnetic characteristics are clearly defined, until we know more of the tectonic and structural history.

At each of the following sites, 10–15, 2.54 cm diameter cores of 6–12 cm length were drilled using a portable two-stroke hand-held machine. Orientation was made with a non-magnetic device and Brunton compass. Magnetization of the volcanics had no effect on the compass orientation. At limestone sites cores were taken from a range of separate beds to sample the complete time-span represented at the outcrop; at the volcanic sites, holes were placed to sample a range of pillows and flows over as large an area as possible to ensure a representative sampling and also to minimize any effects of secular variation.

2.1. Algarinejo limestones and basalts

Along a N–S trending ravine located 1 km west of the town of Algarinejo, submarine volcanic material (pillow lavas and sills) is interbedded within Jurassic red nodular and marly limestones. We sampled several units of pillow lavas (site AB) dated as Aalenian–lower Bajocian (Garcia-Yebra et al., 1972). To reorient samples from the pillow lavas, we assumed the palaeohorizontal to be given by the bedding as measured on the overlying marly limestones. We also sampled a series of limestones some 100 m above the pillow lavas along the road from Loja to Priego (site AC). This section included marly limestones dated as lower–middle Aalenian and red nodular limestones (ammonitico rosso facies) of middle–upper Aalenian age (Garcia-Yebra et al., 1972).

2.2. Montillana

Close to main Granada–Madrid auto route (km 382, 36 km north of Granada) there is an exposure of basic volcanic rocks interbedded with limestones. Studies carried out by García-Dueñas et al. (1976) dated this volcanism as Upper Jurassic. At this locality we sampled basalts (site GMB), interbedded ash flows (site GMC1), and the overlying marly limestones (site GMC2).

2.3. Benalua

We sampled basalts (site BB) and shallow-water limestones (site BC) 3 km north of Colomera in a locality soon to be submerged by an artificial lake. An accurate age of the volcanic rocks is unknown but since it is similar to the general volcanic activity of this area we considered it probably to be of Middle Jurassic age.

2.4. Alamedilla

Different units of pillow lavas were sampled from an exposure of submarine volcanic material located on the road from Guadahortuna to Alamedilla (grouped collectively as site GO). In other localities this volcanism is found in contact with marly limestones from the lower Tithonian and lower Berrasian (Comas et al., 1981, 1986). However, in the sampled area there was no exposed contact between sedimentary rocks and pillow lavas. Therefore to determine the palaeohorizontal we used the following structural elements: (1) major elongation of pillow units, (2) stems of the pillow lavas, (3) columnar jointings, and (4) discontinuities between different lava flows. By averaging measurements of the planes defined by elements (1) and (4) and the normals to (2) and (3), we could define a reasonably consistent bedding orientation.

2.5. Noalejo

At Rio de las Juntas, c. 50 km north of Granada and 1 km southwest of the town of Noalejo, there

TABLE I

Remanent magnetization parameters

Site	Latitude (° N)	Longitude (° W)	Rock type	Before tectonic correction					After tectonic correction				Initial NRM ($A\ m^{-1}$)	Age
				N	D	I	k	α_{95}	D^*	I^*	K^*	α^*_{95}		
This study														
AC	37.34	4.13	Marly limestone	14	74.6	46.2	20.0	9.1	25.8	43.0	28.7	7.6	1.0×10^{-4}–1.2×10^{-2}	Aalenian
AB	37.34	4.13	Pillow lava	6	34.6	34.4	43.4	10.3	22.1	17.4	42.2	10.4	1.3–7.0	Aalen–Bajocian
BB	37.40	3.72	Basalt	11	214.1	26.7	71.5	5.4	35.9	32.4	71.7	5.4	5.0×10^{-1}–9.0	Middle Jurassic
GNV	37.50	3.65	Basalt	10	11.8	53.8	82.7	5.3	32.7	44.7	56.7	6.5	2.5×10^{-2}–2.3×10^{-1}	Middle Jurassic
GNC1	37.50	3.65	Marly limestone	10	9.0	45.7	15.7	12.6	36.3	38.9	15.9	12.5	$1.0\times10^{-4}\times1.2\times10^{-3}$	Middle Jurassic
G0	37.58	3.27	Pillow lava	13	239.4	56.2	32.9	7.3	68.4	11.8	32.8	7.3	5.1×10^{-1}–1.2	Lower Tithonian–lower Berrasian
Unused sites														
BC	37.40	3.72	Limestone	9	15.2	24.6	125.4	4.6	61.6	69.5	150.2	4.2	6.0×10^{-4}–2.0×10^{-3}	Middle Jurassic
	Secondary components?													
GNC2	37.50	3.65	Limestone	6	345.4	32.7	38.0	11.0	79.8	70.2	38.1	11.0	2.4–2.7×10^{-4}	Middle Jurassic
	Uncertain tectonic correction													
GMB	37.49	3.61	Basalt	11	202.5	82.7	42.1	7.1	341.1	3.4	57.9	6.1	3.1×10^{-1}–1.3	Upper Jurassic
	Uncertain tectonic correction													
GMC1	37.49	3.61	Volcanic ash	7	1.7	65.5	29.1	11.4	178.8	44.3	29.2	11.4	3.0×10^{-1}–2.0	Upper Jurassic
	Present-day field direction													
GMC2	37.49	3.61	Limestone	10	127.2	49.9	41.9	7.6	211.5	88.5	41.8	7.6	3.6–6.0×10^{-3}	Upper Jurassic
	Uncertain tectonic correction													

is a large area of submarine volcanics. In one of these sites we sampled volcanic material (site GNV), probably of Middle Jurassic age (García-Dueñas, personal communication, 1986), interbedded white marly limestones (site GNC1), and dark grey marly limestones (site GNC2) close to a tectonic contact with the volcanics. Although in close proximity, the structural corrections for these sites among themselves are very different, and in some places uncertain.

3. Laboratory procedures

The natural remanent magnetization (NRM) of the specimens was measured either with fluxgate spinner magnetometers (Digico and Molspin), or with an ScT cryogenic magnetometer. Measurements were carried out at the palaeomagnetic laboratories of the Institut für Geophysik, ETH Zürich, and the Universidad Complutense, Madrid.

As would be expected from basalts and limestones types, the initial NRM intensities varied from 10^{-4} to 10^{+1} A m^{-1} (Table 1). Several pilot specimens from each site were progressively thermally demagnetized in 25 and 50 °C steps up to 650 °C using a TSD-1 Schonstedt furnace. The bulk magnetic susceptibility was monitored in order to detect formation of new ferromagnetic phases during heating. Another set of pilot samples was demagnetized by the alternating field method in steps of 2.5 mT (up to 20 mT) and 5.0 mT (from 20 to 80 mT) using a Schonstedt demagnetizer or a two-axis tumbler and coil device. Thermomagnetic curves of volcanic rocks were measured by means of a horizontal magnetic balance with a dc field of c. 0.2 T. In addition, gradual acquisition of isothermal remanent magnetization (IRM) to 1.0 T and step-wise thermal demagnetization after application of an additional, perpendicular 0.1 T component were carried out on selected specimens from the limestones using a method of Lowrie and Heller (1982) after slight modification by F. Heller (personal communication, 1985).

For the analysis of palaeomagnetic directions, straight-line segments of demagnetization data were used to calculate palaeomagnetic directions by principal component analysis (Kirschvink, 1980) and linear regression. In a few cases, stable end-points were also determined on vectors which showed directional similarity but did not exhibit straight-line segments on demagnetization diagrams.

4. Magnetic behaviour of the lithologies

4.1. Algarinejo (red nodular and marly limestones)

IRM acquisition, application of a two-component IRM, and subsequent thermal demagnetization of a pilot sample of red nodular limestone (Fig. 2a–c) exhibited several ferromagnetic phases. A stepdown in intensity between 300 and 350 °C (Fig. 2c) was typical in the thermal demagnetization of the NRM and IRM of most samples. This may indicate slight amounts of maghemite or pyrrhotite. However, this component does not appear to have a significant effect on the direction—either NRM or IRM—uncovered during thermal demagnetization (Fig. 2b). The more clearly defined components were:

(1) a phase with medium coercivities and unblocking temperatures around 350–400 °C; analysis of the NRM shows that this component carries a present-day field magnetization and may therefore be a VRM acquired by fine-grained haematite;

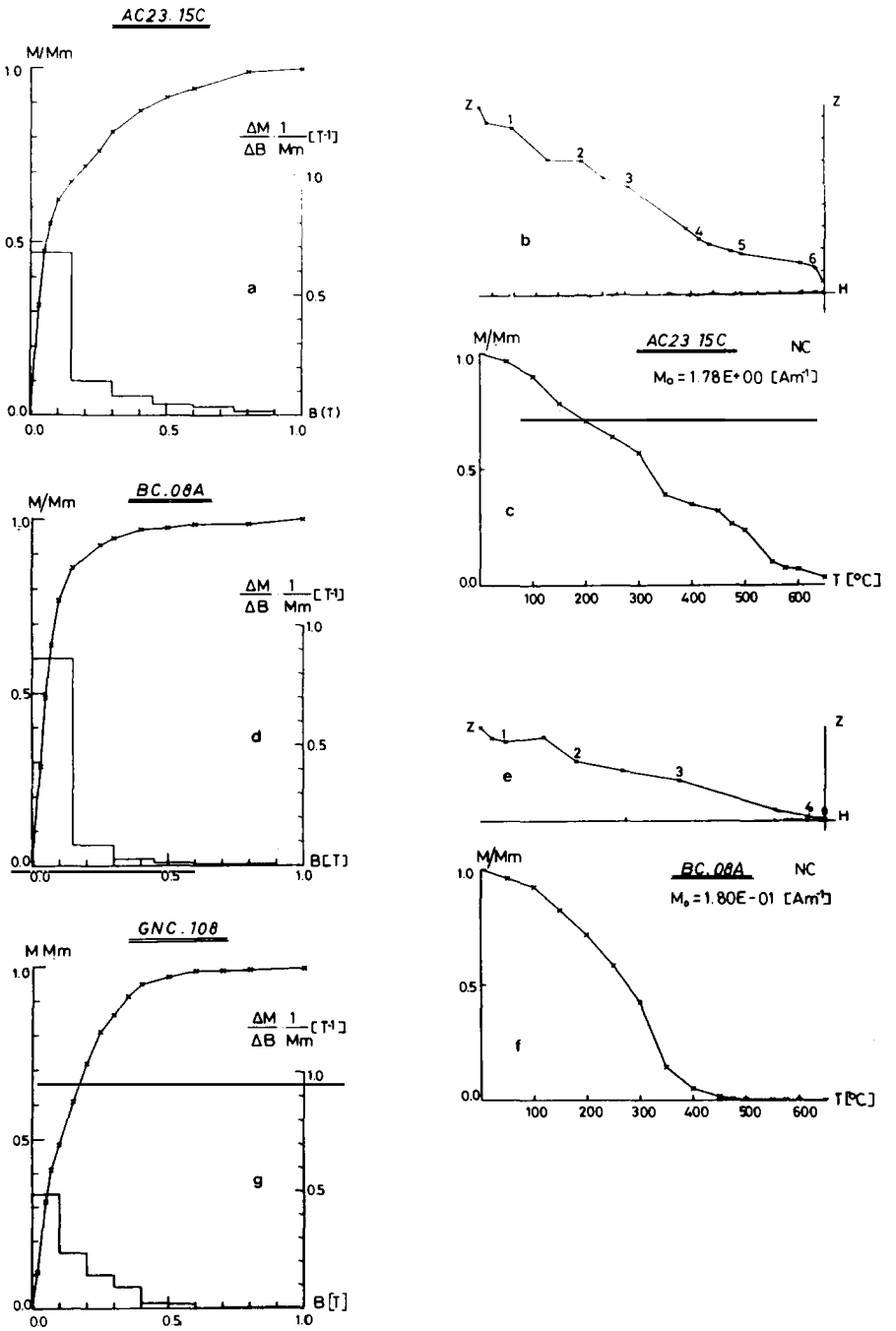
(2) a phase with low coercivities and unblocking temperatures ranging from 350 to 575 °C, probably magnetite;

(3) a phase with high coercivities and unblocking temperatures over 650 °C, most likely due to large-grained (specular) haematite.

Since these pilot results indicated the presence of a high-coercivity phase (haematite), we decided to use thermal treatment for routine demagnetization. The marly limestones presented a magnetic behaviour similar to the red nodular limestones but exhibiting weaker intensity. A Zijderveld diagram of a typical red nodular limestone sample is shown in Fig. 3a; we could clearly isolate two magnetic components:

(1) from 200 to 450 ° C a component with directions close to the present-day field before tectonic correction; and

(2) from 450 to 600 ° C a component probably carried by magnetite which we consider to represent the characteristic magnetization of this site.



A third component appeared in some samples after 600 °C, but the intensity was too weak for accurate determination of the direction.

Published magnetostratigraphic results from the Aalenian are very sparse and poorly determined. This site offers potential for adding more data for this time period. Although the number of samples from site AC is limited and the length of the section is short, we constructed a preliminary magnetostratigraphic profile shown in Fig. 4. These preliminary results appear promising and we intend to make a more extensive sampling to improve the range and resolution of this section.

4.2. Algarinejo (pillow lavas)

Figure 5a shows a thermomagnetic curve of one representative sample (AB.04) of this lithology. This irreversible curve exhibits a ferromagnetic phase with a Curie point of 450–475 °C. Thermal and alternating field techniques were used to analyse the NRM. Both procedures revealed, after the first cleaning steps, only one magnetic component. However, the magnetic behaviour of samples varies from pillow to pillow and we could distinguish two groups:

(1) samples with low (2.5–5 mT) median destructive fields. Direction of the stable component extracted from this range was close to the present-day field; and

(2) samples with median destructive fields in the range 7.5–15.0 mT. The stable direction defined within this range had a reversed magnetization after alternating field treatment above 10 mT. Owing to an (as yet unknown) particularity in the magnetite component of this lithology, we were not able, with our alternating field equipment, to demagnetize these samples above 50 mT. Therefore several samples were thermally demagnetized and stable directions in the range 350–500 °C also revealed reversed polarities. Above 500 °C a new ferromagnetic phase was formed by chemical changes during heating which overwhelmed the NRM direction, at which point we stopped the thermal demagnetization.

4.3. Montillana (limestones and basalts)

In situ directions from samples of the stratified ash lithology (site GMC1) are close to the present-day field direction and the results from this site were therefore rejected for tectonic purposes. Stable characteristic remanent directions of the marly limestones (site GMC2) and basalts (site GMB), are pre-tilting in age, but they were difficult to interpret tectonically. Upon revisiting this locality to check our structural corrections, we observed a 1 m wide zone of asymmetric drag folds near the basalt–limestone contact. This implies a more complex structural situation than originally estimated and we are as yet unable to apply unambiguous tectonic correction procedures on these directions. The tectonic correction used for the volcanics was the strike and dip given by the envelope of the asymmetric drag folds. For the limestones we applied the normal tilt correction as measured directly from the bedding planes.

Fig. 2. Acquisition curves of isothermal remanent magnetization (IRM) with coercivity spectra, and subsequent thermal demagnetization of a two-component IRM. The high-field magnetization (1.0 T) applied first along the axial direction of the sample, then a second, low-field magnetization (0.1 T) along a radial direction. (a), (d), (g) correspond to the IRM acquisition curves. In (b) and (e) the angle between a point on the Z line and the abscissa is the 'inclination', i.e., the angle between the radial (sample co-ordinate = y) direction and magnetization. (e), (f) show the intensity decay during thermal demagnetization of the two IRM components; numbers 1, 2, 3, etc., stand for 100 °C, 200 °C, 300 °C etc. Magnetizations dominated by a low-coercivity phase will plot parallel to the abscissa, those dominated by a high-coercivity phase will plot parallel to the ordinate. The bend at 6 (600 °C) in (b) shows the final demagnetization of all low-coercivity phases, leaving a small but clearly measurable amount of haematite. A low-coercivity component dominates in (e), where the demagnetization was discontinued above 4 (400 °C), the magnetization drop is clearly illustrated in (f). M_0 gives the initial (i.e., high field reduced by the low field) magnetization. Note that the plots are normalized: if (e) was plotted on same scale as (b), the Z path would be so flat (dominated by low-coercivity magnetization) as to be nearly indistinguishable from the abscissa. Sample identifications: 'AC23.15C' red nodular limestone from Algarinejo, 'BC.08A' cream-coloured fine-grained limestone from Benalua, 'GNC.10B' marly limestone from Noalejo.

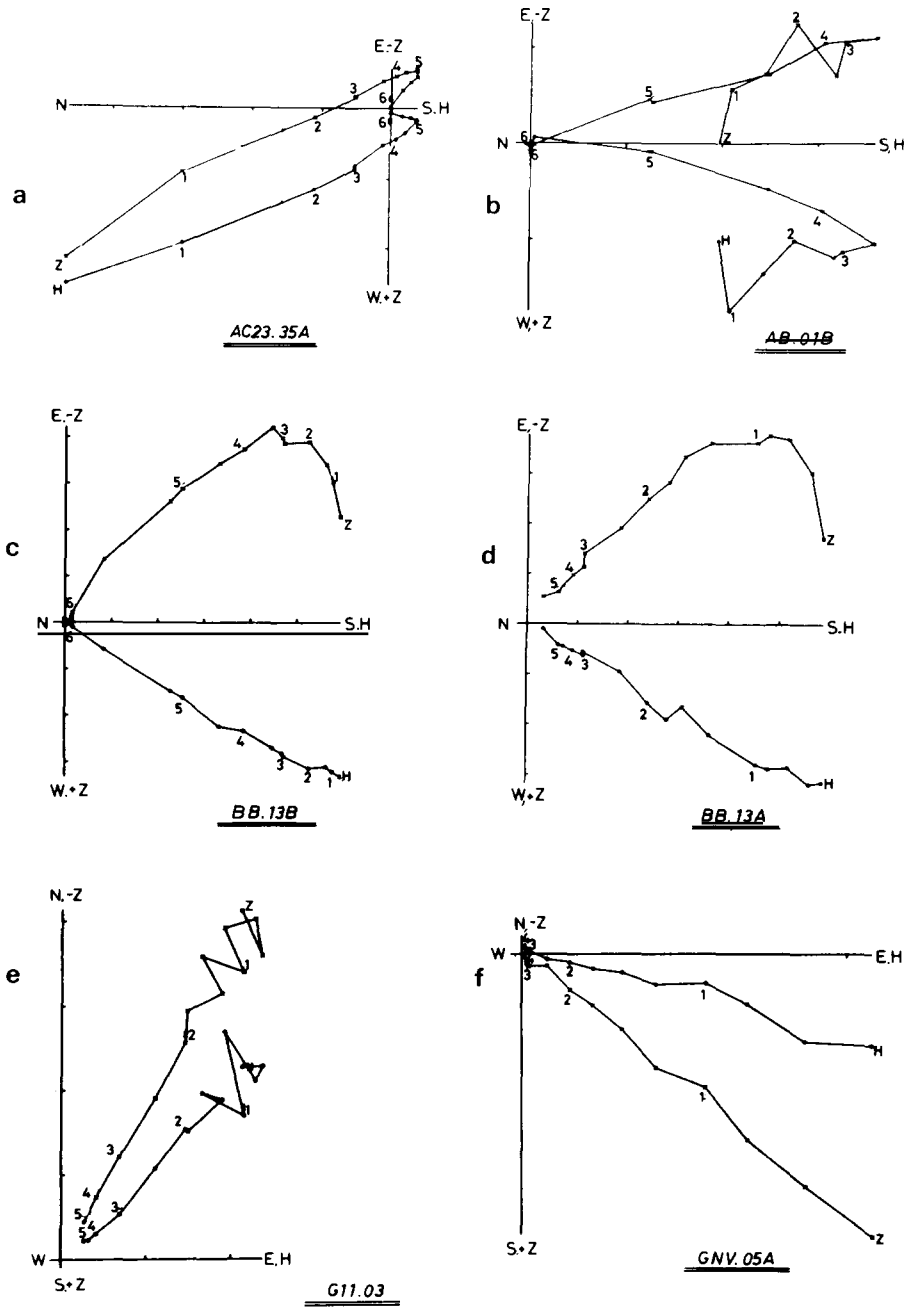


Fig. 3. Orthogonal projection plots of the remanent magnetization path during thermal and alternating field demagnetization of the NRM. In the plots of thermal demagnetization (a–c), the numbers 1, 2, 3, etc., stand for 100 °C, 200 °C, 300 °C, etc., in the plots of alternating field treatment (d–f), the numbers 1, 2, 3, etc., stand for 10, 20, 30 mT, etc. *H*, Horizontal component; *Z*, vertical component. (a) and (b) plotted with tectonic (tilt) correction, (c–e) in situ correction (geographic co-ordinates), and (f) no correction (sample co-ordinates). Letters (AC, AB, BB, G, GNV) identify the sites as referred to in the text and Table I.

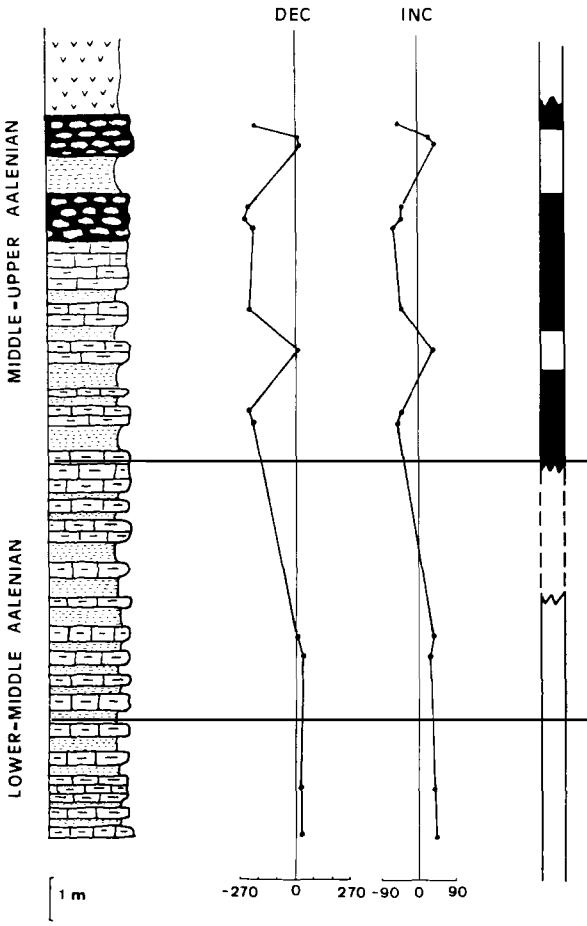


Fig. 4. Palaeomagnetic directions vs. stratigraphic position of the Algarinejo (site AC) section. In the magnetostratigraphic column, black regions are reversed polarity.

4.4. *Benalua (basalts)*

Thermomagnetic curves of basalts from Benalua (Fig. 5b) showed only one magnetic phase with a T_c around 575 °C, implying that the signal is probably carried by nearly pure magnetite. Progressive demagnetization by alternating field and thermal cleaning revealed a stable component above 250–300 °C (Fig. 3c) and/or after alternating field demagnetization to 15–20 mT (Fig. 3d). Unblocking temperatures were in the range of 550–575 °C and the median destructive fields were 15–25 mT. The stable directions from all samples had reversed polarity.

4.5. *Benalua (limestones)*

IRM experiments showed a ferromagnetic phase of low to middle coercivity and unblocking temperatures in the range 300–450 °C (Fig. 2d,f). Thermal demagnetization of the NRM revealed the presence of one stable magnetic component above 100–150 °C. After 300–350 °C the NRM intensity fell below the noise level of the ScT. In spite of this, directions of the stable magnetization were well grouped (precision parameter $k > 100$; Table I). Although the in situ characteristic remanent directions (ChRM) differed significantly from the present-day field, after tectonic correction they had anomalously steep inclinations ($72 \pm 5°$) for the Jurassic palaeogeographic position of the Subbetic Zone. In contrast, ChRM inclinations from the nearby basalts averaged 32°, within the expected range. This anomalous inclination of the limestones may be due to remagnetization at a time of steeper tilt than the present position, or some unrecognized tectonic disturbance. For these reasons, this site was not taken into account in the interpretation. However, it may be noted that the declinations agreed in sense with those from the other sites in this study.

4.6. *Alamedilla (pillow lavas)*

Thermomagnetic curves of this lithology showed two magnetic phases with characteristic Curie points of ~ 375 °C (titanomaghemite?) and 500 °C (titanomagnetite?; Fig. 5c). Both thermal and alternating field cleaning techniques were used for demagnetizing the NRM. Zijderveld plots revealed only one stable component after 20–25 mT alternating field treatment (Fig. 3e) or after heating to 200–250 °C. X-ray diffraction on a magnetic extract exhibited peaks for maghemite, titanomagnetite and haematite. Thermal demagnetization of the NRM showed a decrease in the intensity after heating to 350 °C. However, this decrease of the magnetization did not affect the direction of the NRM, which was the same up to 450–500 °C, the

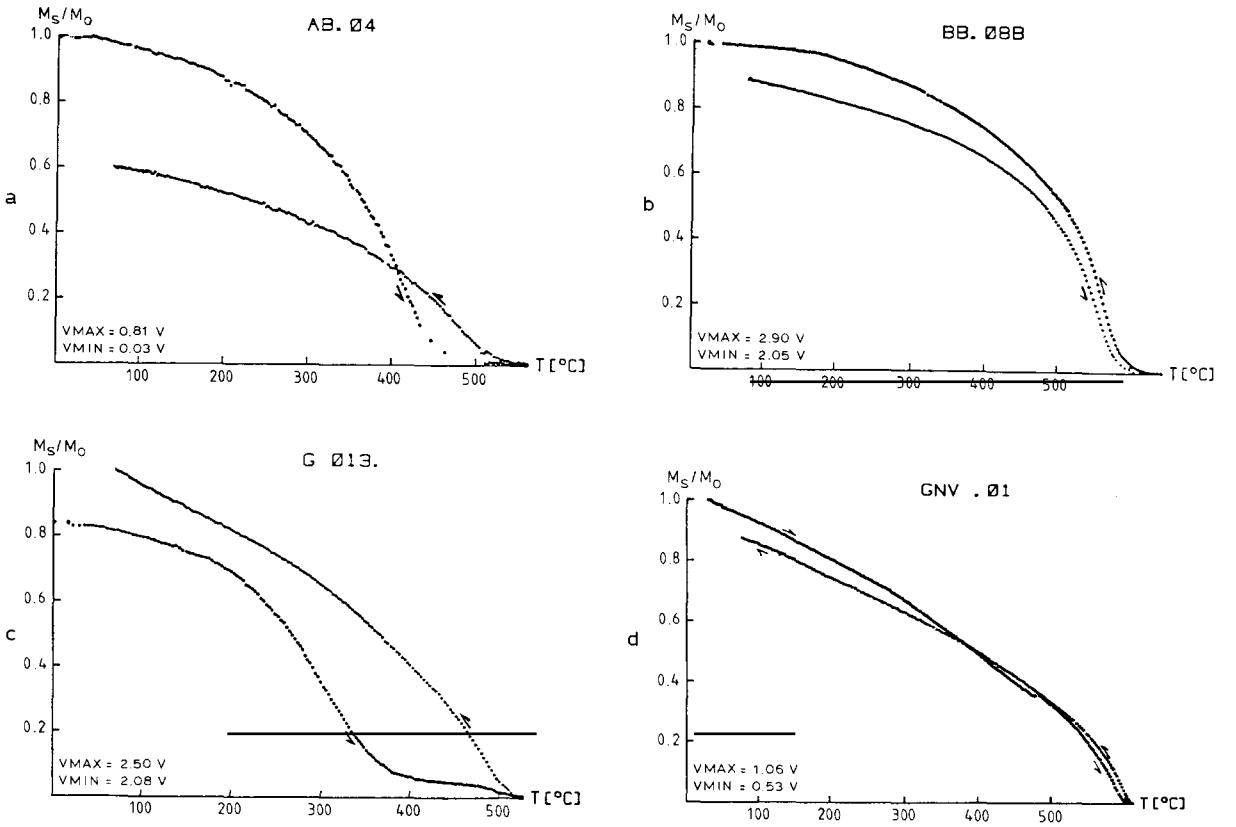


Fig. 5. Thermomagnetic (Curie) curves of samples from submarine volcanic rocks. Specimen identification indicates sampling site.

maximum unblocking temperatures for this lithology.

4.7. Noalejo (volcanics)

A thermomagnetic curve shows a Curie point of c. 585 °C (Fig. 5d), very close to that for pure magnetite. Reflected light microscopy of a polished surface of one sample exhibited lamellae of grey anisotropic material, indicative of high-temperature oxidation to a titanohaematite phase. Zijderveld plots during thermal and alternating field (Fig. 3f) demagnetization showed only one stable component with median destructive fields of 10 mT and unblocking temperatures ranging from 350 to 500 ° C. Despite the evidence for two phases (titanomagnetite and titanohaematite), characteristic remanent directions were similar for both alternating field and thermal demagnetization. This direction had normal polarity.

4.8. Noalejo (white marly limestones)

A typical IRM acquisition curve for this lithology (Fig. 2g) shows the presence of low and middle coercivity phases. Thus both magnetite and haematite may be present. Definition of a stable direction during thermal demagnetization was hampered by creation of new ferromagnetic phases during heating which affected the direction of the magnetization. Unfortunately, as might be expected from the IRM indications for haematite, alternating field treatment was completely ineffective. In spite of this, we were able to isolate a characteristic magnetization using the data from thermal demagnetization before a sharp increase

in the intensity indicated formation of new ferromagnetic phases. The large dispersion of the data ($k = 16$; see Table I) reflects these demagnetization difficulties. However, the inclinations of the stable components have values that are expected for the Jurassic, and ChRM directions yielded mostly reversed polarities. These features support an ancient origin for this magnetization.

4.9. Noalejo (black marly limestones)

Although the magnetic behaviour of this site (GNC2) was adequate for demagnetization and well-defined characteristic directions were obtained (Table I), we have not yet succeeded in correcting for the complex local structural setting.

5. Discussion

5.1. Anomalous inclinations from pillow lavas

All sites of pillow lavas yielded anomalous inclinations (Table I): they were consistently shallower than inclinations measured in nearby limestones and in most cases much shallower (12–17°) than expected for a reasonable estimate of the Jurassic palaeolatitude. This is not a rare finding; similar results have been reported by other workers. For example, Luyendyk et al. (1985) discuss anomalous inclinations in lava flows from Miocene submarine volcanics from California and suggest non-dipole field behaviour, offset dipole, and undetected initial dips as possible causes. We do not think non-dipole, offset dipole, or secular variations (which would affect the declinations also) are valid possibilities in our case. One reason is that in our case all the volcanic rocks have shallower inclinations than the limestones, and in several cases the difference is far too much to be accounted for by non-dipole or secular components. A second reason is that the magnetization of these Jurassic basalts is most probably due to secondary phases (titanomaghemite and titanohaematite by oxidation; Ti-free magnetite by exsolution) acquired during an extensive period of submarine oxidation after cooling although still in the Jurassic. Another possible cause for the shallow declinations is, as reasoned by Kamerling and Luyendyk (1985), that initial (undetected) dips of the pillows could cause an artificial shallowing of the inclination if the pillows are assumed (wrongly) to have been more horizontal originally than they were at the time of acquisition of magnetization.

We have outlined above our criteria for determining the palaeohorizontal in the pillow lavas; in the case of the stems and columnar jointing, the measured dip should be normal to the actual palaeohorizontal at the time of cooling. During this period of submarine oxidation (the presumed time of acquisition of magnetization, long after initial extrusion), the blocks may have had a bedding more steeply inclined (due to thermal contraction, for instance) than the original bedding as indicated by stems and jointing. If this (undetected) dip was northerly, apparent shallow directions would be produced by correction using the stem and jointing dip. However, southerly undetected dips would produce an apparent steepening of the inclination. Again, this explanation runs into the difficulty that all of the volcanics have shallower dips than the limestones. It is conceivable that all the undetected dips (if there were such) happened to be northerly—an interesting implication if true—but we can neither validate nor refute this possibility. Another explanation is that pressure would have forced the ferromagnetic grains to orient shallower than under atmospheric conditions, thus flattening the inclination. We did not measure the anisotropy of magnetic susceptibility and thus cannot judge what effect, if any, this may have had on the NRM. Also, as mentioned above, local structural corrections are in several cases uncertain. In conclusion, since we cannot resolve the problem of the shallow inclinations from the volcanics we do not draw interpretations concerning palaeolatitude from them. The inclinations (39°, 43°) from the two limestone sites we consider reliable lie within the range expected for the Jurassic of southern Spain.

5.2. Palaeomagnetic declinations

The palaeomagnetic results from the central part of the Subbetic Zone presented in this paper indicate a systematic deviation to the east of the

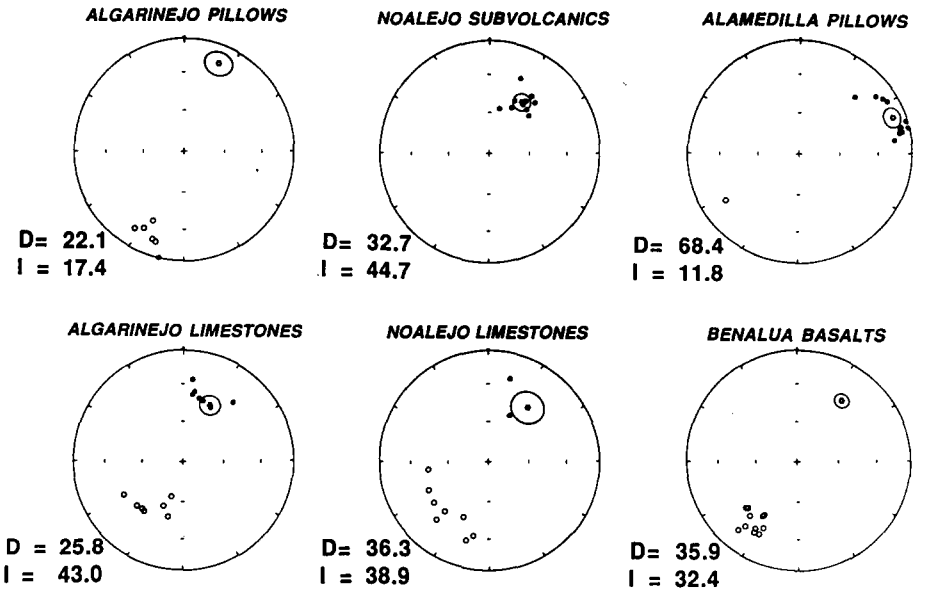


Fig. 6. Stereographic projection plots of the characteristic remanent magnetization vectors after tectonic correction, lower hemisphere equal area projection. ○: upper hemisphere, ●: lower hemisphere. Ellipses show the projection of the 95% confidence circle about the mean direction given by D and I (declination and inclination).

palaeomagnetic declination (Fig. 6 and Table I). The value of this deviation depends on the site and ranges from 20 to 36° in the central part of the studied area to 68° in the easternmost locality. Table II gives the palaeomagnetic results from Jurassic limestones in the central part of the Subbetic Zone obtained by Ogg et al. (1984) and Steiner et al. (1987) and on Cretaceous data from the Prebetic Zone by Vandenberg (1980). Figure 7 illustrates the magnetic declination of the six sites from the present study and five sites from Ogg et al. (1984) and Steiner et al. (1987). In addition, the Jurassic direction for stable Iberia (data from Vandenberg (1980) and Schott et al. (1981)) is given for comparison. The directions from the Subbetic Zone deviate considerably from those from stable Iberia: if the Subbetic Zone was contiguous with stable Iberia during the Cretaceous opening of the Bay of Biscay, they would have been rotated anticlockwise. How can the measured clockwise directions be explained?

The major tectonic alignments in the region around Granada drawn up from Landsat images and field mapping are also depicted in Fig. 7. Solid lines indicate major faults seen on the Landsat images and mapped in the field; broken

TABLE II

Remanent magnetization parameters (other studies)

Site	Latitude (°N)	Longitude (°W)	Rock type	After tectonic correction					Age	Source
				N	D^*	I^*	k^*	α_{95}^*		
Sierra Gorda	37.2	4.1	Limestone	97	323.2	30.0	34.7	2.5	Tithonian–Kimmeridgian	(Ogg et al., 1984)
Carcabuey	37.5	4.3	Limestone	105	26.3	38.3	66.4	1.7	Tithonian–Kimmeridgian	(Ogg et al., 1984)
BJ	37.5	4.3	Limestone	196	53.8	47.0	13.6	2.8	Bathonian–Bajocian	(Steiner et al., 1987)
BC	37.5	4.3	Limestone	112	59.4	42.9	12.1	4.0	Bathonian–Bajocian	(Steiner et al., 1987)
BH	37.3	3.5	Limestone	107	75.8	46.6	8.0	5.2	Bathonian–Bajocian	(Steiner et al., 1987)
SEB	37.6	3.7	Limestone	24	12.4	52.3	?	10.7	Santonian–Campanian	(Vandenberg, 1980)

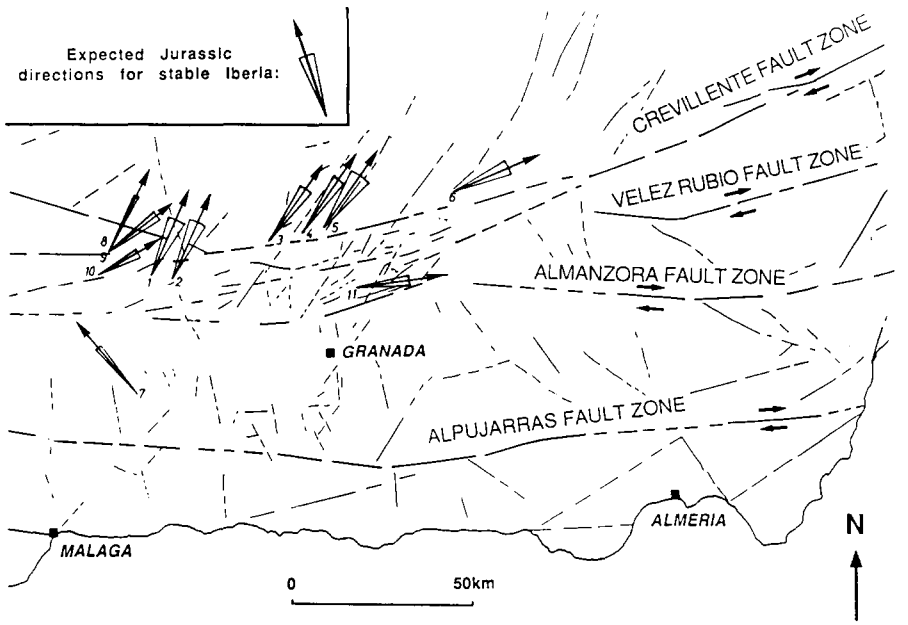


Fig. 7. Map of the central part of the Subbetic Zone showing the sites of the units sampled and the palaeomagnetic directions for each site, the α_{95} is indicated on each arrow. Numbers identify the sites: (1) Algarinejo limestones, (2) Algarinejo pillow lavas (100 m from site 1, separated here for clarity), (3) Benalua basalts, (4) Noalejo undifferentiated submarine volcanics, (5) Noalejo limestones (100 m from site 4, separated here for clarity), (6) Alamedilla pillow basalts, (7) Sierra Gorda limestones (Ogg et al., 1984); (8) Carcabuey limestones (Ogg et al., 1984), (9) Carcabuey limestones (site BJ; Steiner et al., 1987), (10) 'La Fuente' section limestones (site BC; Steiner et al., 1987), (11) Sierra Harana limestones (site BH; Steiner et al., 1987). Inset top left: averaged direction for stable Iberia (data from Vandenberg (1980) and Schott et al. (1981)). Lineations from Landsat images and field observations including unpublished data from R. Vegas and L.-M. Barranco.

lines identify clearly defined linear features seen only in the Landsat images (some of these are partly covered by Neogene sediments). Several major lineations in the Betics correspond to the large E–W trending fault zones; the Crevillente, Velez-Rubio, and Alpujarras fault zones (Fig. 7).

Kinematic models of strike-slip faults and contemporaneous block movements predict rotations of single fault-bounded blocks within shear zones (Freund, 1974), even with opposing sense of rotation. Several recent studies in California (Kamerling and Luyendyk, 1979; Luyendyk et al., 1980, 1985; Hornafius, 1985; Terres and Luyendyk, 1985), in the eastern Mediterranean area (Ron et al., 1984; Kissel et al., 1986), and in the Basin and Range Province (Ron et al., 1987) show large rotations by small blocks within zones of strike-slip faulting.

The palaeomagnetic data from the Subbetic Zone show rotation of blocks having sense and magnitude expected from a model of block rotation in a zone of dextral shear. A schematic diagram (Fig. 8, after Christie-Blick and Biddle (1986)) illustrates the main features of this model. Nicholson et al. (1986a,b) used this model to illustrate rotation of blocks along the San Andreas Fault when the major dextral faults become locked. This may be the situation in the Betics, where E–W movement has been locked up due to changes in the overall stress regime in the western Mediterranean owing to the Africa–Eurasia plate convergence. Note that the sense of movement along the NE–SW trending faults will be sinistral strike slip. This is the situation found in the Alahama de Murcia, Northern Betic, Palomares, and Carboneras Faults in the Betics (Bousquet and Montenat, 1974; Bousquet and Philip, 1976; Montenat et al., 1987) and the Nekkor, Jebha, and Mellila Faults in the Alboran Sea and Morocco (Leblanc and Olivier, 1984; Bouillon et al., 1986). Moreover, the

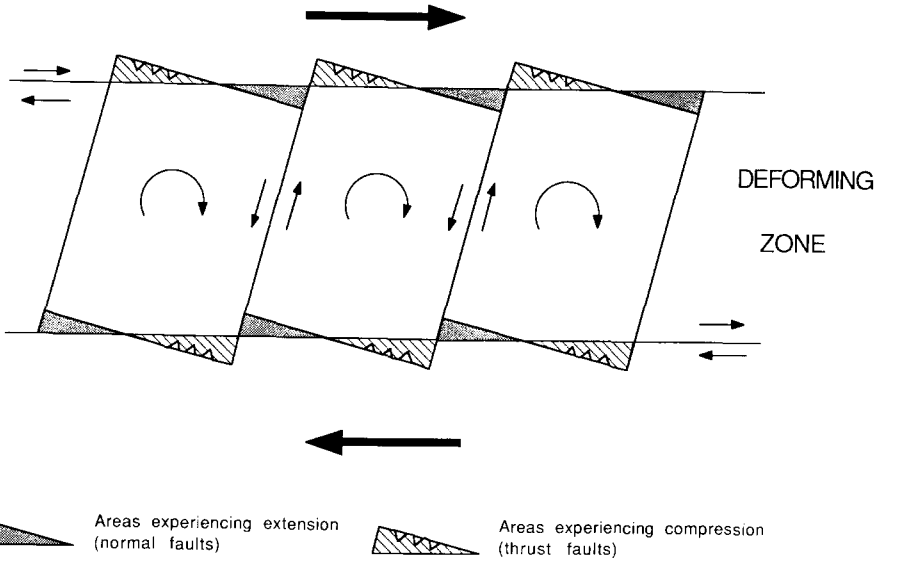


Fig. 8. Model showing how blocks could rotate clockwise in a dextral shear zone. Note that the NE–SW faults are sinistral, and that both reverse and normal faulting could be expected as the blocks rotate. Taken from Christie-Blick and Biddle (1986), who refer to Dibblee (1977) and Nicholson et al. (1986a,b) as sources. Nicholson et al. (1986b) display this model as a possible mechanism for rotating blocks when the master faults (in their case, the San Andreas Fault in California; in our case, the Crevillente Fault or North Betic Fault) have been locked, preventing major strike slip along main trace.

model indicates how reverse and normal faulting may occur in areas experiencing compression and extension as a consequence of the block rotation.

The tectonic situation in the Betics can be seen in the larger picture as a zone, caught between Africa and the Iberian Meseta, of diffuse distributed deformation in the sense described by McKenzie and Jackson (1983, 1986) for the Aegean area. This situation in the western Mediterranean is depicted in Fig. 9(a,b). Most plate kinematic reconstructions (Olivet et al., 1982) shows left-lateral shear between north Africa and Iberia between 180 and 80 Ma. The models of Dewey et al. (1973) and Biju-Duval et al. (1977) show dextral movement between 80 and 38 Ma followed by N–S compression. Other models (Tapponier, 1977; Olivet, 1978; Livermore and Smith, 1985) do not indicate any dextral 'kink'. There is thus some uncertainty in the relative sense of movement since the Upper Cretaceous.

Palaeomagnetic data, in contrast to determination of polarity (sense) of fault movement using instantaneous features (e.g., slickensides on fault scarps), can show the overall polarity integrating many small movements over a large time-span. Thus they more accurately represent the true, overall sense of movement. At present, we do not have enough data (both palaeomagnetic and structural) to resolve the uncertainty about the sense of movement along major E–W faults in southern Spain. In fact, it appears most likely that both dextral and sinistral movements have occurred, but at different times. Relying on field evidence, many authors (Paquet, 1974; Araña and Vegas, 1974; Van de Fliert et al., 1980; De Smet, 1984; Leblanc and Olivier, 1984) have described the Subbetics and Betics as a zone of dextral megashear during the Tertiary. The Crevillente Fault Zone (CFZ in Fig. 9a) cuts Subbetic units and, although still active today, is presumed to have moved mostly between the middle and upper Miocene (Leblanc and Olivier, 1984).

The palaeomagnetic results presented in this study reflect the sum of tectonic movements since the Jurassic. At the present, we have no data from Cretaceous or Tertiary rocks with which to confine the timing of movements implied by the clockwise declinations. Thus the declinations may

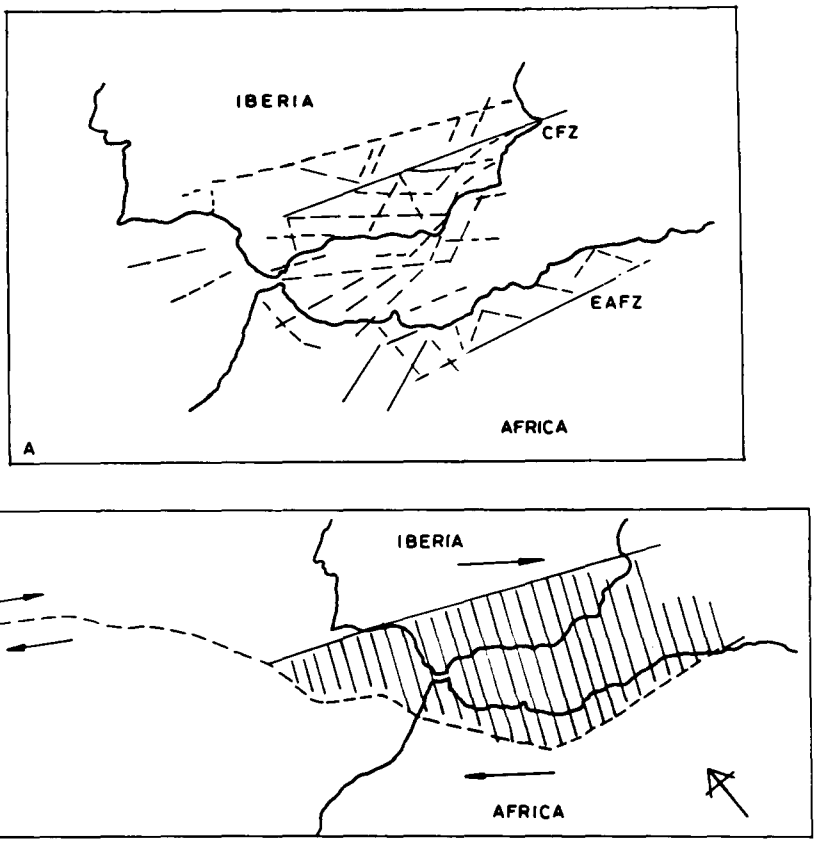


Fig. 9. (A) Extent of the zone of diffuse deformation between Africa and Iberia with major strike-slip faults and secondary transverse faults. CFZ: Crevillente Fault Zone; EAFZ: El Asnam Fault Zone (after R. Vegas and L.-M. Barranco, unpublished data). (B) Geodynamic setting of the broad zone of intermediate deformation between the Iberian and African plates. Open arrow: sense of convergence between Iberia and Africa for the Oligo-Miocene; solid arrows: direction of shear induced by the Neogene movement of Africa relative to Iberia (Eurasia). AGTZ: Azores–Gibraltar transform zone.

be a product of rotations produced during transtension in the Mesozoic, movement of Iberia in the Cretaceous, or transpression since the Upper Cretaceous. However, considering the violence of the Alpine orogeny in this region, we tend to favour Tertiary tectonic events as most likely to explain the bulk of the declination deviations.

Although the model of clockwise rotation of blocks in a dextral megashear zone can explain the major elements in this region, numerous complexities remain. One example is provided by the palaeomagnetic data of Ogg et al. (1984) from the Sierra Gorda Range, West of Granada (site 7, Fig. 7). This site exhibits westerly declinations, which contrast with the consistent easterly declinations from all other sites included in our survey. Although it represents only one site (new sampling in this area is under way) and is therefore tentative, it does present an apparent contradiction to our model. There are, however, several reasons to account for anticlockwise rotations within the same general model. Sierra Gorda is located very close to the North Betic Fault which was probably right-lateral in the Oligo-Miocene, but left-lateral (albeit of lesser importance) during the late Miocene. In addition, some of the more complex models of block rotations discussed by Garfunkel and Ron (1985) exhibit simultaneous but opposite rotations of adjacent blocks. According to these authors, the sense of movement depends on the

original orientation of the faults bounding the blocks relative to the overall stress regime. From a similar point of view, Hornafius (1985) discusses anticlockwise movements in a dextral strike-slip zone of the eastern Transverse Ranges of California. In the Subbetic Zone we need more data both to confirm that the Sierra Gorda direction is representative, and to define the geometry and sense of movement of other blocks in this region. We hope that future work integrating palaeomagnetics, seismicity, and structural geology will help to contribute to a more detailed understanding of this region.

6. Summary

Palaeomagnetic results from both Jurassic volcanics and limestones indicate eastward declinations considerably deviating from the expected directions for stable Iberia. The sense of these declinations supports a model of rotation of crustal blocks within a zone of intense dextral shear active during the later stages of the plate convergence between Africa and Iberia. Further palaeomagnetic investigations are in progress to help define more precisely the timing and extent of these rotations.

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